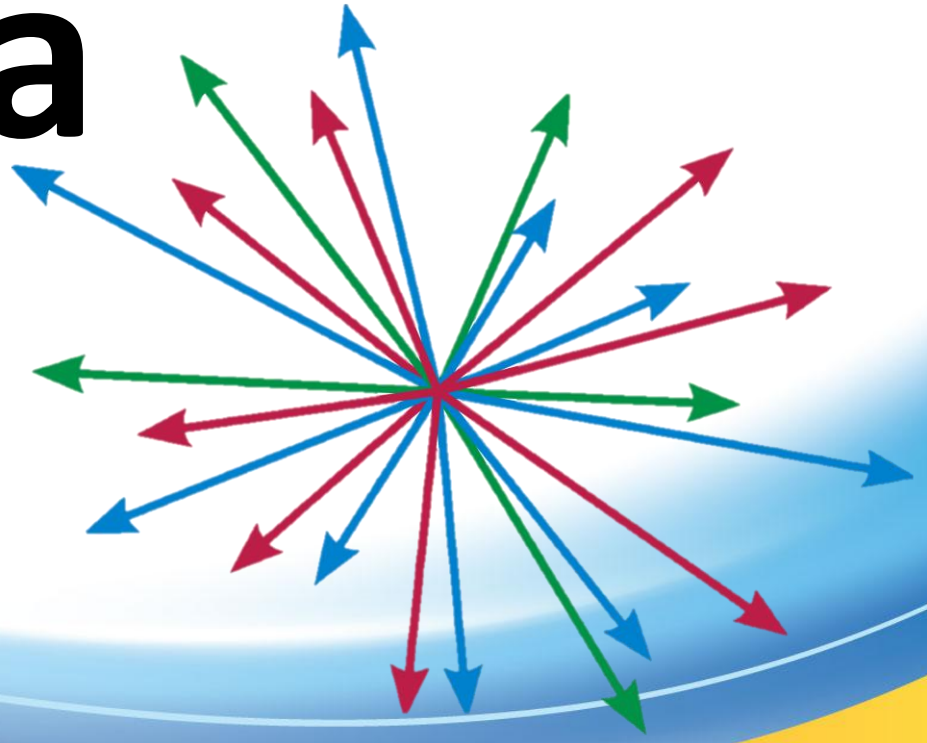
Undergraduate Course in Mathematics

Linear Algebra



Topic: Orthogonality in IPS

Conducted By

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Orthogonality

Abstraction of 90°

Inner Product

Let V be a vector space over the field F. An inner product on V is a function that takes each ordered pair $(\vec{u}, \vec{v})$ of elements of V to a number $\langle \vec{u}, \vec{v} \rangle \in F$ and has the following properties:

Positivity: $\langle \vec{u}, \vec{u} \rangle \geq 0$ for all $\vec{u} \in V$;

Definiteness: $\langle \vec{u}, \vec{u} \rangle = 0$ if and only if $\vec{u} = 0$;

Additivity in first slot: $\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$ for all $\vec{u}, \vec{v}, \vec{w} \in V$;

Homogeneity in first slot: $\langle \lambda \vec{u}, \vec{v} \rangle = \lambda \langle \vec{u}, \vec{v} \rangle$ for all $\lambda \in F$ and for all $\vec{u}, \vec{v} \in V$;

Conjugate Symmetry: $\langle \vec{u}, \vec{v} \rangle = \overline{\langle \vec{v}, \vec{u} \rangle}$ for all $\vec{u}, \vec{v} \in V$.

Orthogonality

Let V be a vector space over the field F and $\left\langle \overrightarrow{\quad}, \overrightarrow{\quad} \right\rangle$ be an inner product. Two vectors $\vec{u}$ and $\vec{v}$ are called orthogonal if

$$\langle \vec{u}, \vec{v} \rangle = 0$$

Orthogonality of Vectors

Orthogonal Vectors in $\mathbb{R}^n$

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \right\rangle = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n = 0$$

$$= \sum_{i=1}^{n} x_i y_i \quad \checkmark$$

for orthogonality

Show that the vectors **(1,−1,0), (1,1,−2)** are orthogonal with respect to standard inner product in $\mathbb{R}^3$.

$$\langle (1,-1,0), (1,1,-2) \rangle$$

$$= 1 \cdot 1 + (-1) \cdot 1 + 0 \cdot (-2)$$

$$= 0$$

$\therefore$ $(1,-1,0)$ and $(1,1,-2)$ are orthogonal.

Show that the vectors $(2+i, 1-3i)$ and $(10, 1+7i)$ are orthogonal with respect to standard inner product in $\mathbb{C}^2$.

$$\langle (2+i, 1-3i), (10, 1+7i) \rangle$$

$$= (2+i)\cdot\overline{10} + (1-3i)\cdot\overline{(1+7i)}$$

$$= (2+i)\cdot 10 + (1-3i)\cdot(1-7i)$$

$$= 20 + 10i + 1 - 7i - 3i - 21 = 0$$

$$\overline{a+ib} = a-ib$$

$\therefore (2+i, 1-3i)$ and $(10, 1+7i)$

are orthogonal.

Orthogonality of functions

Orthogonal functions

$$\langle f(x), g(x) \rangle = \int_a^b f(x) \cdot g(x)\, dx = 0$$

for orthogonality

Consider the inner product

$$\langle f(x), g(x)\rangle = \int_{-1}^{1} f(x)g(x)dx$$

Show that the functions $\underbrace{3x^2 - 1}$ and $\underbrace{5x^3 - 3x}$ are orthogonal.

$$\langle 3x^2 - 1, 5x^3 - 3x\rangle = \int_{-1}^{1} (3x^2-1)(5x^3-3x)\,dx$$

$$= \int_{-1}^{1} (15x^5 - 9x^3 - 5x^3 + 3x)\,dx$$

$$= \left[15\frac{x^6}{6} - \frac{9}{4}x^4 - \frac{5}{4}x^4 + \frac{3}{2}x^2 \right]_{-1}^{1}$$

$$= \left(\frac{15}{6} - \frac{9}{4} - \frac{5}{4} + \frac{3}{2} \right) - \left(\frac{15}{6} - \frac{9}{4} + \frac{5}{4} + \frac{3}{2} \right)$$

$$= 0$$

$\therefore$ $3x^2 - 1$ and $5x^3 - 3x$ are orthogonal.

Consider the inner product

$$\langle f(x), g(x)\rangle = \int_{-\pi}^{\pi} f(x)g(x)dx$$

Show that the functions $\sin(x)$ and $\cos(2x)$ are orthogonal.

$$\langle \sin x, \cos 2x \rangle = \int_{-\pi}^{\pi} \sin x \cdot \cos 2x \, dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi} 2\sin(x)\cos(2x)\, dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi}\left[\sin(x+2x) + \sin(x-2x)\right]dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi}(\sin 3x)\,dx + \frac{1}{2}\int_{-\pi}^{\pi}\sin(-x)\,dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi}\sin(3x)\,dx - \frac{1}{2}\int_{-\pi}^{\pi}\sin x\,dx$$

$$= \frac{1}{2}\left[\frac{-\cos 3x}{3}\right]_{-\pi}^{\pi} - \frac{1}{2}\left[-\cos x\right]_{-\pi}^{\pi}$$

$$= \frac{1}{2}\left[\frac{-\cos 3\pi}{3} - \frac{-\cos(-3\pi)}{3}\right] - \frac{1}{2}\left[(-\cos\pi) - (-\cos(-\pi))\right]$$

$$= \frac{1}{2}\left[\frac{-(-1)}{3} - \frac{-(-1)}{3}\right] - \frac{1}{2}\left[-(-1) - (-1)(-1)\right]$$

$$= 0.$$

$\therefore$ $\sin x$ and $\cos 2x$ are orthogonal.

w.r.t the given inner product.

$$\begin{cases} 2\sin A\cos B = \sin(A+B) + \sin(A-B) \\ 2\sin A\sin B = \cos(A-B) - \cos(A+B) \\ 2\cos A\cos B = \cos(A+B) + \cos(A-B) \end{cases}$$

$$\begin{cases} 2\sin^2 A = 1 - \cos 2A \\ 2\cos^2 A = 1 + \cos 2A \end{cases}$$

$$\sin(-x) = -\sin x$$

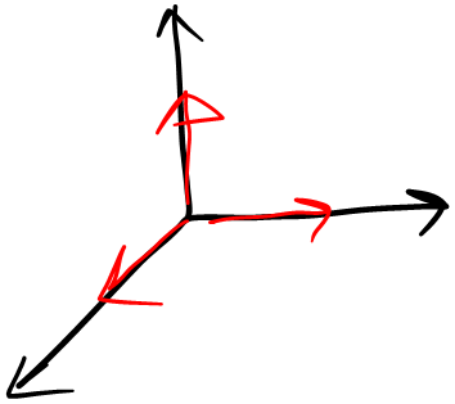
$$\cos(-x) = \cos x$$

Orthogonal and Orthonormal Set

Orthogonal Vectors

Let V be a vector space over the field F and $\langle \overrightarrow{\quad}, \overrightarrow{\quad} \rangle$ be an inner product. A set of vectors $S = \{\overrightarrow{v_1}, \overrightarrow{v_2}, \overrightarrow{v_3}, \dots, \overrightarrow{v_k}\}$ is called orthogonal if

$$\langle \overrightarrow{v_m}, \overrightarrow{v_n} \rangle = 0 \text{ for all } m \neq n$$



Orthonormal Vectors

Let V be a vector space over the field F and $\langle \overrightarrow{\quad}, \overrightarrow{\quad} \rangle$ be an inner product. A set of vectors $S = \{\overrightarrow{v_1}, \overrightarrow{v_2}, \overrightarrow{v_3}, \dots, \overrightarrow{v_k}\}$ is called orthonormal if

$$\left.\begin{aligned} \langle \overrightarrow{v_m}, \overrightarrow{v_n} \rangle &= 0 \text{ for all } m \neq n \\ \langle \overrightarrow{v_m}, \overrightarrow{v_n} \rangle &= 1 \text{ for all } m = n \end{aligned}\right\}$$

(Orthogonal) + (length = 1)

How to convert an Orthogonal set to an Orthonormal set

$$\frac{V_k}{\|\vec{V}_k\|} \rightarrow \text{normalize}$$

Show that

$$B = \left\{(0,1,0), \left(-\frac{4}{5}, 0, \frac{3}{5}\right), \left(\frac{3}{5}, 0, \frac{4}{5}\right)\right\}$$

Is an orthonormal Set.

$$\therefore \left\langle (0,1,0), \left(-\frac{4}{5}, 0, \frac{3}{5}\right)\right\rangle = 0 + 0 + 0 \quad \checkmark$$

$$\therefore \left\langle (0,1,0), \left(\frac{3}{5}, 0, \frac{4}{5}\right)\right\rangle = 0 + 0 + 0 \quad \checkmark$$

$$\therefore \left\langle \left(-\frac{4}{5}, 0, \frac{3}{5}\right), \left(\frac{3}{5}, 0, \frac{4}{5}\right)\right\rangle = -\frac{12}{25} + 0 + \frac{12}{25} = 0$$

$\therefore$ B is orthogonal.

$$\langle (0,1,0), (0,1,0) \rangle = 0 + 1 + 0 = 1 \quad \checkmark$$

$$\langle (-\tfrac{4}{5}, 0, \tfrac{3}{5}), (-\tfrac{4}{5}, 0, \tfrac{3}{5}) \rangle = \frac{16}{25} + 0 + \frac{9}{25} = 1 \quad \checkmark$$

$$\langle (\tfrac{3}{5}, 0, \tfrac{4}{5}), (\tfrac{3}{5}, 0, \tfrac{4}{5}) \rangle = \frac{9}{25} + 0 + \frac{16}{25} = 1 \quad \checkmark$$

$\therefore$ B is orthonormal.

Show that

$$\mathbf{B} = \{(\mathbf{1}, \mathbf{1}, \mathbf{1}), (1,-1,0), (1,1,-2)\}$$

Is an orthogonal set but not orthonormal. Convert this set to an orthonormal set.

$\therefore \langle (1,1,1), (1,-1,0) \rangle = 1-1+0 = 0$ ✓

$\therefore \langle (1,1,1), (1,1,-2) \rangle = 1+1-2 = 0$ ✓

$\therefore \langle (1,-1,0), (1,1,-2) \rangle = 1-1+0 = 0$ ✓

$\therefore$ B is orthogonal ✓

$$\therefore \langle (1,1,1), (1,1,1) \rangle = 3 \neq 1$$

$\Rightarrow$ B is not orthonormal.

Conversion

$$\|\vec{v}_1\| = \sqrt{\langle (1,1,1),(1,1,1)\rangle}$$

$$= \sqrt{3}.$$

$$\therefore \vec{e}_1 = \frac{\vec{v}_1}{\sqrt{3}} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right).$$

$$\|\vec{v}_2\| = \sqrt{\langle \vec{v}_2, \vec{v}_2 \rangle} = \sqrt{\langle (1,-1,0), (1,-1,0) \rangle}$$

$$= \sqrt{2}$$

$$\vec{e}_2 = \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0 \right).$$

$$\|\vec{v}_3\| = \sqrt{\langle \vec{v}_3, \vec{v}_3 \rangle}$$

$$= \sqrt{\langle (1,1,-2), (1,1,-2) \rangle} = \sqrt{6}$$

$$\vec{e}_3 = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}} \right).$$

$\therefore$ orthonormal set.

$$\left\{ \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right), \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0 \right), \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}} \right) \right\}$$

$$\begin{cases} 2\sin A\cos B = \sin(A+B) + \sin(A-B) \\ 2\sin A\sin B = \cos(A-B) - \cos(A+B) \\ 2\cos A\cos B = \cos(A+B) + \cos(A-B) \end{cases}$$

$$\begin{cases} 2\sin^2 A = 1 - \cos 2A \\ 2\cos^2 A = 1 + \cos 2A \end{cases}$$

$$\sin(-x) = -\sin x$$

$$\cos(-x) = \cos x$$

$$\sin(n\pi) = 0 \qquad n \in \mathbb{Z}$$

$$\cos(n\pi) = (-1)^n \qquad n \in \mathbb{Z}.$$

$$\sin(\text{integer} \cdot \pi) = 0$$

$$\cos(\text{integer} \cdot \pi) = (-1)^{\text{integer}}$$

Prove that for $m, n \in \mathbb{Z}$

$$\int_{-\pi}^{\pi} \sin(nx)\cos(mx)\,dx = 0$$

$$= \frac{1}{2}\int_{-\pi}^{\pi} 2 \cdot \sin(nx)\cos(mx)\,dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi} \left[\sin(nx+mx) + \sin(nx-mx)\right] dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi} \sin\left((n+m)x\right) dx + \frac{1}{2}\int_{-\pi}^{\pi} \sin\left((n-m)x\right) dx$$

$$= \frac{1}{2}\left[\frac{-\cos\left((n+m)x\right)}{n+m}\right]_{-\pi}^{\pi} + \frac{1}{2}\left[\frac{-\cos\left((n-m)x\right)}{n-m}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{2}\left[\frac{\cos\left((n+m)x\right)}{n+m}\right]_{\pi}^{-\pi} + \frac{1}{2}\left[\frac{\cos\left((n-m)x\right)}{n-m}\right]_{\pi}^{-\pi}$$

$$= \frac{1}{2}\left[\frac{\cos\left((n+m)\pi\right)}{n+m} - \frac{\cos\left((n+m)\pi\right)}{n+m}\right]$$

$$+ \frac{1}{2}\left[\frac{\cos\left(-(n-m)\pi\right)}{n-m} - \frac{\cos\left((n-m)\pi\right)}{n-m}\right]$$

$$= \frac{1}{2}\left[\frac{\cos(n+m)\pi}{n+m} - \frac{\cos(n+m)\pi}{n+m}\right] + \frac{1}{2}\left[\frac{\cos(n-m)\pi}{n-m} - \frac{\cos(n-m)\pi}{n-m}\right]$$

$= 0.$

$\therefore$ $\sin(nx)$ and $\cos(mx)$ are orthogonal.

Prove that for $m, n \in \mathbb{Z}$

$$\int_{-\pi}^{\pi} \sin(nx)\sin(mx)dx = \begin{cases} 0, & m \neq n \\ \pi, & m = n \end{cases}$$

Case - 1 $\quad n \neq m.$

$$\int_{-\pi}^{\pi} \sin(nx)\,\sin(mx)\,dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi} 2\cdot\sin(nx)\cdot\sin(mx)\,dx$$

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$$= \frac{1}{2}\int_{-\pi}^{\pi}\left[\cos(nx-mx) - \cos(nx+mx)\right]dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi}\cos((n-m)x)\,dx - \frac{1}{2}\int_{-\pi}^{\pi}\cos((n+m)x)\,dx$$

$$= \frac{1}{2}\left[\frac{\sin((n-m)x)}{n-m}\right]_{-\pi}^{\pi} - \frac{1}{2}\left[\frac{\sin((n+m)x)}{n+m}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{2}\left[\frac{\sin((n-m)\pi)}{n-m} - \frac{\sin((n-m)(-\pi))}{n-m}\right]$$

$$- \frac{1}{2}\left[\frac{\sin((n+m)\pi)}{n+m} - \frac{\sin((n+m)(-\pi))}{n+m}\right]$$

$$= \frac{1}{2}\left[\frac{\sin(n-m)\pi}{n-m} + \frac{\sin(n-m)\pi}{n-m}\right] - \frac{1}{2}\left[\frac{\sin(n+m)\pi}{n+m} + \frac{\sin(n+m)\pi}{n+m}\right]$$

$$= \frac{\sin\left((n-m)\pi\right)}{n-m} - \frac{\sin\left((n+m)\pi\right)}{n+m}$$

$$= \frac{0}{n-m} - \frac{0}{n+m} = 0 \quad \checkmark$$

Case-2 $\quad n = m$

$$\int_{-\pi}^{\pi} \sin(nx) \cdot \sin(nx)\, dx$$

$$= \frac{1}{2} \int_{-\pi}^{\pi} 2 \cdot \sin^2(nx)\, dx$$

$$= \frac{1}{2} \int_{-\pi}^{\pi} \left(1 - \cos(2nx) \right) dx$$

$$= \frac{1}{2} \int_{-\pi}^{\pi} 1 \, dx - \frac{1}{2} \int_{-\pi}^{\pi} \cos(2nx) \, dx$$

$$= \frac{1}{2} \left[x \right]_{-\pi}^{\pi} - \frac{1}{2} \left[\frac{\sin(2nx)}{2n} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{2}\left[\pi - (-\pi)\right] - \frac{1}{2}\left[\frac{\sin(2n)\pi}{2n} - \frac{\sin(2n)(-\pi)}{2n}\right]$$

$$= \frac{1}{2}\cdot 2\pi - \frac{1}{2}\left[\frac{0}{2n} - \frac{0}{2n}\right]$$

$$= \pi.$$

Show that the set of the function $(\cos kx)$, $k = 1,2,3,\ldots$ is orthogonal over the interval $(-\pi, \pi)$. Also check whether they are orthonormal or not. If not, construct an orthonormal set of function from the above set of function.

<u>claim</u> $\int_{-\pi}^{\pi} \cos(mx)\cos(nx)\,dx = 0$ for $m \neq n$.

$$= \frac{1}{2}\int_{-\pi}^{\pi} 2\cos(mx)\cdot\cos(nx)\,dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi}\left[\cos(mx+nx) + \cos(mx-nx)\right]dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi}\cos((m+n)x)\,dx + \frac{1}{2}\int_{-\pi}^{\pi}\cos((m-n)x)\,dx$$

$$= \frac{1}{2}\left[\frac{\sin((m+n)x)}{m+n}\right]_{-\pi}^{\pi} + \frac{1}{2}\left[\frac{\sin((m-n)x)}{m-n}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{2}\left[\frac{\sin((m+n)\pi)}{m+n} - \frac{\sin((m+n)(-\pi))}{m+n}\right]$$

$$\frac{1}{2}\left[\frac{\sin((m-n)\pi)}{m-n} - \frac{\sin((m-n)(-\pi))}{m-n}\right]$$

$$= 0.$$

$\therefore \{\cos(kx) : k \in \mathbb{N}\}$ is orthogonal.

Ho

$$\langle \cos kx, \cos kx \rangle = \int_{-\pi}^{\pi} \cos kx \cdot \cos kx \, dx$$

$$= \frac{1}{2} \int_{-\pi}^{\pi} 2\cos^2 kx \, dx$$

$$= \frac{1}{2} \int_{-\pi}^{\pi} \left[1 + \cos(2kx) \right] dx$$

$$= \frac{1}{2}\left[x\right]_{-\pi}^{\pi} + \frac{1}{2}\left[\frac{\sin(2kx)}{2k}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{2}\left(\pi-(-\pi)\right) + \frac{1}{2}\left[\frac{\sin(2k\pi)}{2k} - \frac{\sin(2k(-\pi))}{2k}\right]$$

$$= \pi.$$

$$\|\cos(kx)\| = \sqrt{\pi}.$$

$$\therefore \left\{ \frac{\cos(kx)}{\sqrt{\pi}} : k \in \mathbb{N} \right\} \text{ is orthonormal.}$$

$$= \left\{ \frac{\cos x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, \frac{\cos 3x}{\sqrt{\pi}}, \ldots \right\}$$

Thank you!